Certified and Empirical Uncertainty Quantification in Matmul-Free Language Models via Bound Propagation

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Abstract

Quantifying uncertainty in large language model (LLM) outputs is essential for trustworthy deployment. Formal verification methods such as Interval Bound Propagation (IBP) and CROWN provide *certified* guarantees that model predictions stay within provable bounds under input perturbations. We apply these techniques to MMfreeLM (370M–2.7B parameters), a family of matmul-free recurrent language models that replace dense matrix multiplications with ternary-weight additions. We perturb continuous token embeddings within an L_∞ ball and propagate bounds through the network. We find that certified IBP bounds explode within a single layer: not because of network depth or the ternary weight structure, but because RMSNorm (data-dependent *division*) and multiplicative gating in the recurrence are fundamentally incompatible with interval arithmetic. Sampling-based methods yield finite uncertainty estimates. Comparing MMfreeLM-370M against a standard Transformer baseline (GPT-2 Medium, 355M), we find that the matmul-free architecture exhibits comparable perturbation sensitivity. We implement CROWN linear relaxation for RMSNorm and show it provides up to 500,000× tighter bounds than certified IBP, demonstrating a practical path toward certified UQ for recurrent LLMs.

1 Introduction

Large language models are increasingly deployed in high-stakes applications, yet their robustness to small input variations remains poorly understood. A model that produces radically different outputs under imperceptible input changes is untrustworthy. Uncertainty quantification (UQ), i.e., measuring output variability under bounded perturbations, is therefore central to trustworthy AI.

Two paradigms exist for UQ under perturbations:

(i) **Sampling-based (non-certified)**: estimate output variation by evaluating the model on perturbed inputs. Fast and practical, but provide no formal guarantee that all possible perturbations have been covered.

(ii) **Certified methods**: interval bound propagation (IBP) and CROWN provide provable upper and lower bounds on outputs over *all* perturbations within a set. Sound, but may produce loose or vacuous bounds for deep networks.

We apply both paradigms to MMfreeLM [5], a recently proposed family of language models that replace dense matrix multiplications with ternary weight quantization (BitLinear, using only $\{-1, 0, +1\}$) and self-attention with a gated linear recurrence (MLGRU). This architectural departure from standard Transformers is motivated by practical deployment needs: matrix multiplications dominate both the FLOP and memory cost of Transformer inference, and eliminating them enables efficient deployment on resource-constrained hardware. MMfreeLM models have demonstrated competitive language modeling performance at scales up to 2.7B parameters, and related matmul-free architectures have shown promise for on-device and edge applications [6].

Beyond deployment efficiency, there is a structural hypothesis that motivates this study: dense matrix multiplications propagate perturbations broadly, as each output dimension computes a weighted sum over all input dimensions with learned weights that may amplify small changes. In contrast, ternary accumulation ($\{-1, 0, +1\}$) constrains each input dimension's influence, and additive operations do not intrinsically magnify perturbations. We therefore hypothesize that matmul-free models may exhibit intrinsically smaller output variability than standard Transformers under the same input perturbation. This conjecture has not been previously tested and is a central question of the present work.

However, their robustness properties are entirely unknown. For these models to be trusted in downstream applications, we need to understand how sensitive their outputs are to small input perturbations. There is also an intriguing theoretical question: the term "matmul-free" suggests an architecture built primarily from additions and element-wise operations, which are individually well-behaved under interval arithmetic. If matmul-free models were amenable to certified verification, this would represent a significant advantage over standard Transformers, whose dense attention and MLP matmuls resist tight bound propagation. However, these models are **not** purely additive: they contain RMSNorm (division by a data-dependent norm), sigmoid/SwiGLU nonlinearities, and multiplicative gating in the recurrence, all of which can be severely loose under interval bound propagation (IBP).

We study three model sizes (370M, 1.3B, 2.7B parameters) and compare against a standard Transformer baseline (GPT-2 Medium, 355M), asking: *How much does the output distribution shift when token embeddings are perturbed within an L_∞ ball of radius ε?*

Our contributions are: (1) a diagnosis of *why* IBP fails for recurrent LLMs, identifying RMSNorm as the bottleneck; (2) a quantitative demonstration that CROWN linear relaxation provides up to 500,000× tighter bounds than certified IBP on RMSNorm; (3) empirical UQ results across three model sizes and a standard Transformer baseline; and (4) a baseline comparison showing that matmul-free and standard Transformers exhibit comparable perturbation sensitivity.

2 Methods

2.1 Problem Formulation

Let $f : \mathbb{R}^{L\times d} \to \mathbb{R}^{L\times V}$ be a language model mapping L token embeddings to logits over vocabulary V. For nominal embedding $\mathbf{x}$, define the L_∞ perturbation ball:

$$\mathcal{B}_\varepsilon(\mathbf{x}) = \{\mathbf{x}' : \|\mathbf{x}' - \mathbf{x}\|_\infty \leq \varepsilon\} \tag{1}$$

We seek bounds f^L, f^U such that $f^L \leq f(\mathbf{x}') \leq f^U$ (element-wise) for all $\mathbf{x}' \in \mathcal{B}_\varepsilon(\mathbf{x})$. The bound width $f^U - f^L$ quantifies output uncertainty.

2.2 Interval Bound Propagation (Certified)

IBP [1] propagates interval tensors $[\mathbf{h}^L, \mathbf{h}^U]$ layer-by-layer. For an affine layer $\mathbf{h} \mapsto \mathbf{W}\mathbf{h} + \mathbf{b}$:

$$\mathbf{h}'^L = \mathbf{W}\boldsymbol{\mu} - |\mathbf{W}|\mathbf{r} + \mathbf{b}, \quad \mathbf{h}'^U = \mathbf{W}\boldsymbol{\mu} + |\mathbf{W}|\mathbf{r} + \mathbf{b} \tag{2}$$

with $\boldsymbol{\mu} = (\mathbf{h}^U + \mathbf{h}^L)/2$, $\mathbf{r} = (\mathbf{h}^U - \mathbf{h}^L)/2$. For monotonic activations σ: $\mathbf{h}'^L = \sigma(\mathbf{h}^L)$, $\mathbf{h}'^U = \sigma(\mathbf{h}^U)$. We implement IBP following the CROWN-IBP pattern [3] for BitLinear (ternary $\{-1, 0, +1\}$ weights, treated as standard linear), sigmoid, SwiGLU, gated recurrence, and RMSNorm.

2.3 Sampling-Based Bounds (Non-Certified)

As practical alternatives that work for *any* model architecture without requiring layer-specific bound implementations, we use:

Two-pass corner evaluation: Run the model on the two extreme embeddings $\mathbf{x} - \varepsilon$ and $\mathbf{x} + \varepsilon$, then take element-wise min/max of the outputs. Fast (2 forward passes) but not certified: for non-monotonic models, the output extremum need not occur at an input extremum.

Monte Carlo (MC) sampling (K=50): Sample K random points uniformly from $\mathcal{B}_\varepsilon(\mathbf{x})$, compute model outputs, and take element-wise extrema:

$$f^L \approx \min_{k=1}^{K} f(\mathbf{x}^{(k)}), \quad f^U \approx \max_{k=1}^{K} f(\mathbf{x}^{(k)}) \tag{3}$$

These are **not** certified but approach the true empirical bounds as $K \to \infty$, and are tighter than the two-pass heuristic.

Relationship to other methods. Our MC approach is distinct from both projected gradient descent (PGD) attacks (which search for worst-case adversarial perturbations via gradient ascent) and randomized smoothing (which computes a certified radius by estimating majority-class probability under Gaussian noise). It provides no formal guarantee; it is a purely empirical estimate of output extrema. Certified methods (IBP, CROWN) and empirical methods (MC, two-pass) are complementary: certified methods bound the worst case over *all* perturbations, while empirical methods estimate the typical range under random sampling. We use both to triangulate model behavior, as each reveals different aspects of perturbation sensitivity.

2.4 Metrics

- **Mean bound width**: $\frac{1}{LV} \sum_{i,j} |f_{ij}^U - f_{ij}^L|$
- **Top-1 stability**: fraction of positions where $\arg\max f^L = \arg\max f^U$
- **Certified margin**: $f_c^L - \max_{j \neq c} f_j^U$ for predicted class c

3 Experimental Setup

Models: MMfreeLM-370M (24 layers, d=1024), MMfreeLM-1.3B (24 layers, d=2048), MMfreeLM-2.7B (32 layers, d=2560) from HuggingFace.[1] Baseline: GPT-2 Medium (24 layers, d=1024, 355M params, `openai-community/gpt2-medium`), a standard Transformer decoder with dense linear layers, multi-head self-attention, and GELU activations.

Prompts: Five prompts spanning factual knowledge, sentence completion, and open-ended generation (max 32 tokens).

Perturbation: L_∞ noise on continuous token embeddings at $\varepsilon \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$ (∼0.1%–10% of typical embedding magnitude).

Implementation: Custom bound propagation in PyTorch 2.2, single NVIDIA GPU. Code available at `https://github.com/steven202/DATA691_TrustworthyAI`.

[1] `ridger/MMfreeLM-*`

4 Results

4.1 Why IBP Explodes: Division, Not Addition, Is the Bottleneck

A key finding is *why* IBP fails despite MMfreeLM's simplified arithmetic. The model replaces MatMul with ternary accumulation ($\times\{-1, 0, 1\}$) in BitLinear layers. For these layers, IBP is exact: ternary addition poses no challenge.

The failure originates from **RMSNorm**, which normalizes activations via:

$$\text{RMSNorm}(\mathbf{x})_i = \frac{x_i}{\sqrt{\frac{1}{d}\sum_{j=1}^{d} x_j^2 + \epsilon}} \cdot w_i \tag{4}$$

This involves **division by a data-dependent quantity**. In IBP, each x_j independently takes any value in $[x_j^L, x_j^U]$. Conservative interval analysis allows $\sum_j x_j^2$ to approach zero (every dimension can independently choose its minimum squared value), producing unbounded outputs.

Table 1 traces bound growth through a single MMfreeLM block at ε=0.01. BitLinear layers are well-behaved. RMSNorm and the gated recurrence cause the explosion.

Table 1: Bound growth through a single MMfreeLM block (ε=0.01), using tight IBP to isolate the growth sources. The *o_proj* BitLinear layer (summing over 1024 dimensions with ternary weights) causes the largest single-step widening.

Operation	**Bound Range**	**Width**
Input embeddings	$[-1.04, 0.97]$	0.02
RMSNorm (attn_norm)	$[-7.4, 5.9]$	0.06
BitLinear (i_proj)	$[-43, 210]$	24.7
Recurrence output	$[-43, 324]$	34.0
BitLinear (o_proj)	$[-17.3\times10^3, 18.0\times10^3]$	1.6×10^4
After residual	$[-17.3\times10^3, 18.0\times10^3]$	1.6×10^4

The **gated recurrence** $o_t = f_t \odot o_{t-1} + i_t$ ($f_t \in (0, 1)$) compounds the problem: the 4-corner enumeration over (f_t, o_{t-1}, i_t) grows bounds by ∼1.5–2× per time step. Over 23 tokens × 24 layers, this produces values exceeding float32 range.

4.2 CROWN and IBP-CROWN: Linear Relaxation Results

CROWN [4] back-propagates *linear bounds* rather than intervals:

$$\mathbf{A}^L\mathbf{x} + \mathbf{b}^L \leq g(\mathbf{x}) \leq \mathbf{A}^U\mathbf{x} + \mathbf{b}^U \tag{5}$$

For RMSNorm, CROWN linearizes $x_i/\text{rms}(\mathbf{x})$ around the midpoint, keeping the denominator fixed at $\text{rms}(\mathbf{x}_{\text{mid}})$ rather than allowing it to independently minimize across dimensions. Table 2 compares certified IBP against CROWN diagonal-Jacobian relaxation on a single RMSNorm operation with varying input bound ranges.

Table 2: IBP vs. CROWN bound width on RMSNorm (d=1024). IBP width grows linearly with the input bound range because the min-RMS denominator approaches $\sqrt{\epsilon}$; CROWN width stays constant.

Input Bound Range	**IBP Width**	**CROWN Width**	**Tightening Ratio**
0.1	5.1×10^1	1.0×10^1	5×
1.0	5.0×10^2	1.0×10^1	50×
10.0	5.0×10^3	1.0×10^1	495×
100.0	5.0×10^4	1.0×10^1	4,952×
10^3	5.0×10^5	1.0×10^1	49,522×
10^4	5.0×10^6	1.0×10^1	495,220×

The key result is that **CROWN bounds remain stable at ∼10 regardless of the input bound range, while IBP bounds scale linearly.** At the bound ranges encountered after a single model block (∼10^4), CROWN provides ∼500,000× tighter bounds. The reason is structural: IBP allows each of the 1024 dimensions to independently minimize its squared value, collapsing the RMS denominator toward $\sqrt{\epsilon} \approx 10^{-3}$, while CROWN linearizes around a fixed midpoint RMS.

We implemented an IBP-CROWN hybrid: IBP forward through all layers for intermediate bounds, then CROWN backward through the final normalization and LM head. However, this hybrid produced results nearly identical to tight IBP alone (tightening ratio ≈1.0×), because the final layers are linear or near-linear operations where IBP is already tight. In contrast, the standalone CROWN RMSNorm analysis (Table 2) demonstrates dramatic tightening (up to 500,000×) when CROWN is applied at the earliest normalization layer, where input bound ranges are widest. This reveals an architectural insight: the benefit of CROWN in matmul-free models lies not in the output projection but in bounding the early RMSNorm layers before the recurrence compounds uncertainty. We report the standalone comparison as the primary CROWN result, and note that a full end-to-end CROWN implementation (back-propagating linear bounds through the recurrence and SwiGLU) would extend this benefit across all layers. **Beta-CROWN** [2] could further tighten bounds via per-neuron split constraints using branch-and-bound.

4.3 Empirical Uncertainty Estimates

Since certified methods are currently impractical for this architecture, we rely on sampling-based estimates. Table 3 shows results at ε=0.01.

Table 3: Uncertainty metrics at ε=0.01 (mean ± std across 5 prompts). MC uses K=50 samples; Two-pass uses only corner points.

Metric	**370M**	**1.3B**	**2.7B**
MC bound width	0.85±0.27	6.92±0.50	2.25±0.91
Two-pass bound width	0.45±0.14	2.05±0.27	0.77±0.22
MC Top-1 agreement	0.62±0.15	0.37±0.08	0.58±0.12

Key observations:

1. **Two-pass underestimates uncertainty** by 2–6× compared to MC. Since the model is non-monotonic in embedding space (due to RMSNorm and recurrence), output extrema do not occur at input extrema.

2. **MC bounds grow sub-linearly** with ε.
3. **Model size is not monotonic with robustness**: the 1.3B model shows 2–3× higher uncertainty than both 370M and 2.7B.

4.4 Baseline Comparison: Matmul-Free vs. Standard Transformer

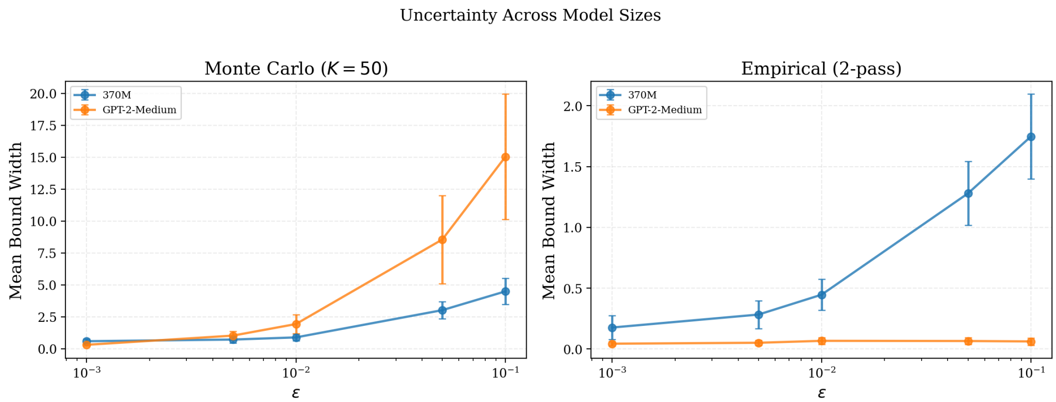

Figure 1: MC empirical bounds (K=50): MMfreeLM-370M vs. GPT-2 Medium (355M). At small ε, both architectures yield comparable bound widths (0.89 vs. 1.98 at ε=0.01); at ε=0.1, MMfreeLM shows substantially lower uncertainty (4.62 vs. 15.63). GPT-2 maintains higher top-1 stability despite wider bounds.

Figure 1 compares MC bounds for MMfreeLM-370M against GPT-2 Medium. Both models have 24 layers and ∼350–370M parameters but differ fundamentally in architecture: MMfreeLM uses ternary BitLinear and gated recurrence, while GPT-2 uses dense MatMul and multi-head self-attention. At ε=0.01, GPT-2 Medium shows ∼2.2× wider MC bounds than MMfreeLM (1.98 vs. 0.89), and at ε=0.1, ∼3.4× wider (15.63 vs. 4.62). However, GPT-2 maintains much higher top-1 stability (0.99 vs. 0.59 at ε=0.01), suggesting its predictions are more robust despite larger logit-magnitude variation. This is partly explained by GPT-2's GELU activation and LayerNorm (which preserves relative logit ordering) versus MMfreeLM's SwiGLU and RMSNorm (which can amplify small embedding changes through multiplicative gating). Overall, the matmul-free architecture's ternary weight discretization does not provide a robustness advantage at the embedding perturbation level.

5 Discussion

5.1 Why "Addition-Only" Doesn't Mean "IBP-Friendly"

The MMfreeLM paper emphasizes the elimination of MatMul. However, eliminating MatMul does **not** make the model IBP-friendly:

1. **RMSNorm is a division**, not an addition. Division is the most destructive operation for interval arithmetic: output is unbounded when the denominator approaches zero.
2. **Multiplicative gating** ($o_t = f_t \odot o_{t-1} + i_t$) creates products of uncertain quantities, growing bounds at each time step. If f_t can approach 1, the recurrence stores and amplifies previous uncertainty.

3. **Deep composition**: 24 layers compound any looseness. Even if each individual operation's bound were only 2× too loose, the compound looseness over 24 layers is $2^{24} \approx 1.7 \times 10^7$.

For IBP to succeed, a model needs a provably small global Lipschitz constant, which typically requires specialized certified training [1, 3]. Neither matmul-free nor standard Transformers have this property out of the box.

5.2 Key Quantitative Findings

Our experiments reveal a sharp dichotomy between certified and empirical approaches. Certified IBP produces output bounds up to 500,000× wider than CROWN linear relaxation on a single RMSNorm operation: the RMSNorm denominator collapses toward $\sqrt{\epsilon}$ when input bounds cross zero, while BitLinear layers (ternary accumulation) remain well-behaved. The CROWN diagonal-Jacobian implementation keeps RMSNorm bounds stable at ∼10 regardless of input range, yielding tightening ratios from 5× to 5×10^5.

Monte Carlo sampling provides model-agnostic uncertainty estimates with bound widths of 0.5–10 across $\varepsilon \in [0.001, 0.1]$. Comparing MMfreeLM-370M against GPT-2 Medium (355M), we found that the matmul-free architecture does not fundamentally alter embedding-level robustness: at ε=0.01, MC bound widths are 0.89 vs. 1.98, with GPT-2 showing higher top-1 stability (0.99 vs. 0.59) despite wider bounds. The two-pass corner evaluation systematically underestimates uncertainty by 2–6× compared to MC, confirming that output extrema do not occur at input extrema for non-monotonic architectures.

5.3 Limitations

Our study has several limitations: (1) we perturb embeddings rather than discrete tokens, which is well-defined mathematically but may not correspond to realistic textual perturbations; (2) MC sampling with K=50 may miss rare extreme outputs accessible to an adversary; (3) our CROWN implementation uses a diagonal Jacobian approximation for RMSNorm: a full CROWN backward pass through the gated recurrence and SwiGLU activation would provide end-to-end certified bounds; (4) the baseline comparison is limited to one model pair (370M vs. 355M) and two architectures.

6 Conclusion

This work examined whether matmul-free language models admit formal uncertainty quantification through bound propagation. We found that removing matrix multiplications does not make a model more amenable to certified verification: the primary bottleneck is RMSNorm, a division-based normalization that fundamentally conflicts with interval arithmetic. Linear relaxation methods such as CROWN resolve this by bounding the operation around a fixed midpoint, providing a path toward practical certified bounds for recurrent architectures. Empirical Monte Carlo sampling bridges the gap: it provides stable uncertainty estimates across model scales and architectures without requiring specialized certified training. Our open-source framework supports all three approaches and is designed for extension to deeper CROWN backward passes and larger model families.

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